Number Systems and Data Units

Source: Open technical documentation and standard references (public knowledge, paraphrased)

This document gives a short overview of Number Systems and Data Units.

The binary system uses base 2 with only the digits 0 and 1. The hexadecimal system uses base 16, with digits 0 to 9 and letters A to F. The octal system uses base 8.

The decimal number 10 is written as 1010 in binary. One byte equals 8 bits. One kilobyte is commonly taken as 1024 bytes, and one megabyte as 1024 kilobytes.

ASCII is a 7-bit character encoding that represents 128 characters. UTF-8 is a Unicode encoding that uses one to four bytes per character and can represent text from all languages.

Note: This is a simplified educational summary written in our own words from public information. It is not legal, financial or medical advice. Verify details with the official source before relying on them.